

Worked examples for Lecture 22: Free oscillations

David Al-Attar

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These examples accompany Lecture 22. They are intended as practice in the concrete calculations that underlie the lecture, and are less demanding than the questions on the problem sets. Numerical results were checked with the scripts in the python directory.

Example 1: A two-degree-of-freedom analogue. Two particles of mass m and $2m$ move along a line. The first is attached to a fixed wall by a spring of stiffness k , the two particles are joined by a spring of stiffness $2k$, and the second is attached to a second fixed wall by a spring of stiffness $2k$. Let $u_1(t)$ and $u_2(t)$ be the displacements from equilibrium, and let external forces $f_1(t)$ and $f_2(t)$ act on the particles. (a) Write the equations of motion in the form $\mathbf{M}\ddot{\mathbf{u}} + \mathbf{K}\mathbf{u} = \mathbf{f}$. (b) Find the eigenfrequencies and eigenvectors, show that the eigenvectors are orthogonal with respect to $\mathbf{M}$, and normalise them so that $\mathbf{s}_k^T \mathbf{M} \mathbf{s}_k = 1$. (c) Show that in the frequency domain $\tilde{\mathbf{u}} = \sum_k \frac{\mathbf{s}_k (\mathbf{s}_k^T \tilde{\mathbf{f}})}{\omega_k^2 - \omega^2}$. (d) For the step force $\mathbf{f}(t) = (0, F_0)^T H(t)$, with H the Heaviside function, obtain the time-domain solution and check its long-time average against the static solution.

Solution. (a) The spring forces on the first particle are $-ku_1$ from the wall spring and $-2k(u_1 - u_2)$ from the coupling spring, while the second particle feels $-2k(u_2 - u_1)$ and $-2ku_2$. Newton's second law therefore gives

$$\begin{pmatrix} m & 0 \\ 0 & 2m \end{pmatrix} \begin{pmatrix} \ddot{u}_1 \\ \ddot{u}_2 \end{pmatrix} + k \begin{pmatrix} 3 & -2 \\ -2 & 4 \end{pmatrix} \begin{pmatrix} u_1 \\ u_2 \end{pmatrix} = \begin{pmatrix} f_1 \\ f_2 \end{pmatrix}, \quad (1)$$

which defines $\mathbf{M}$ and $\mathbf{K}$. Both matrices are symmetric, and both are positive definite. This is the finite-dimensional counterpart of the weak form $\langle \mathbf{w} | P | \ddot{\mathbf{u}} \rangle + \langle \mathbf{w} | H | \mathbf{u} \rangle = \langle \mathbf{w} | \mathbf{f} \rangle$ in the lecture, with the correspondences $\langle \mathbf{w} | P | \mathbf{u} \rangle \leftrightarrow \mathbf{w}^T \mathbf{M} \mathbf{u}$, $\langle \mathbf{w} | H | \mathbf{u} \rangle \leftrightarrow \mathbf{w}^T \mathbf{K} \mathbf{u}$ and $\langle \mathbf{w} | \mathbf{f} \rangle \leftrightarrow \mathbf{w}^T \mathbf{f}$. Since everything is real we can work with real vectors, and the symmetry of $\mathbf{M}$ and $\mathbf{K}$ is the analogue of the self-adjointness of the kinetic and potential energy forms.

(b) Substituting $\mathbf{u} = \mathbf{s} e^{i\omega t}$ with $\mathbf{f} = \mathbf{0}$ gives the generalised eigenvalue problem $\mathbf{K} \mathbf{s} = \omega^2 \mathbf{M} \mathbf{s}$. Non-trivial solutions require

$$\det(\mathbf{K} - \omega^2 \mathbf{M}) = (3k - m\omega^2)(4k - 2m\omega^2) - 4k^2 = 2(m\omega^2 - k)(m\omega^2 - 4k) = 0, \quad (2)$$

so that the squared eigenfrequencies are $\omega_1^2 = k/m$ and $\omega_2^2 = 4k/m$. For ω_1^2 the matrix $\mathbf{K} - \omega_1^2 \mathbf{M}$ has rows proportional to $(1, -1)$, giving $\mathbf{s}_1 \propto (1, 1)^T$; for ω_2^2 the rows are

proportional to $(1, 2)$, giving $\mathbf{s}_2 \propto (2, -1)^T$. The mass-weighted inner product of these vectors is $m(1 \times 2) + 2m(1 \times (-1)) = 0$, so they are $\mathbf{M}$ -orthogonal even though they are not orthogonal in the usual sense. This is a general fact: from $\mathbf{s}_2^T \mathbf{K} \mathbf{s}_1 = \omega_1^2 \mathbf{s}_2^T \mathbf{M} \mathbf{s}_1$ and $\mathbf{s}_1^T \mathbf{K} \mathbf{s}_2 = \omega_2^2 \mathbf{s}_1^T \mathbf{M} \mathbf{s}_2$, the symmetry of $\mathbf{K}$ and $\mathbf{M}$ shows that the left-hand sides are equal, and hence $(\omega_1^2 - \omega_2^2) \mathbf{s}_1^T \mathbf{M} \mathbf{s}_2 = 0$, exactly as in the lecture. Normalising, we take

$$\mathbf{s}_1 = \frac{1}{\sqrt{3m}} \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \quad \mathbf{s}_2 = \frac{1}{\sqrt{6m}} \begin{pmatrix} 2 \\ -1 \end{pmatrix}, \quad \mathbf{s}_j^T \mathbf{M} \mathbf{s}_k = \delta_{jk}. \quad (3)$$

Because the two eigenvectors are linearly independent, any vector can be expanded as $\mathbf{u} = \sum_k (\mathbf{s}_k^T \mathbf{M} \mathbf{u}) \mathbf{s}_k$, which is the two-dimensional version of the completeness relation in the lecture.

(c) With the transform convention of the lecture, $\tilde{\mathbf{u}}(\omega) = \int_0^\infty \mathbf{u}(t) e^{-i\omega t} dt$, and zero initial conditions, two integrations by parts give $\tilde{\ddot{\mathbf{u}}} = -\omega^2 \tilde{\mathbf{u}}$, and so $-\omega^2 \mathbf{M} \tilde{\mathbf{u}} + \mathbf{K} \tilde{\mathbf{u}} = \tilde{\mathbf{f}}$. Writing $\tilde{\mathbf{u}} = \sum_k c_k \mathbf{s}_k$, using $\mathbf{K} \mathbf{s}_k = \omega_k^2 \mathbf{M} \mathbf{s}_k$, and premultiplying by $\mathbf{s}_j^T$, we find that $(\omega_j^2 - \omega^2) c_j = \mathbf{s}_j^T \tilde{\mathbf{f}}$, and so

$$\tilde{\mathbf{u}}(\omega) = \sum_{k=1}^2 \frac{\mathbf{s}_k (\mathbf{s}_k^T \tilde{\mathbf{f}})}{\omega_k^2 - \omega^2}, \quad (4)$$

which is the eigenfunction expansion solution of the lecture with the bra-kets replaced by matrix products. The resonances at $\omega = \pm\omega_k$ are evident.

(d) The time-domain solution is easiest to obtain by projecting the equations of motion rather than by inverting eq. (4). Let $q_k(t) = \mathbf{s}_k^T \mathbf{M} \mathbf{u}(t)$ be the modal coordinates, so that $\mathbf{u} = \sum_k q_k \mathbf{s}_k$. Premultiplying eq. (1) by $\mathbf{s}_k^T$, and using the transposed eigenvalue equation $\mathbf{s}_k^T \mathbf{K} = \omega_k^2 \mathbf{s}_k^T \mathbf{M}$, we obtain the uncoupled oscillator equations

$$\ddot{q}_k + \omega_k^2 q_k = \mathbf{s}_k^T \mathbf{f}(t), \quad q_k(0) = \dot{q}_k(0) = 0. \quad (5)$$

For the step force $\mathbf{s}_k^T \mathbf{f} = F_k H(t)$ with $F_k = \mathbf{s}_k^T(0, F_0)^T$ constant, and one checks by differentiation that the solution for $t > 0$ is

$$q_k(t) = F_k \frac{1 - \cos \omega_k t}{\omega_k^2}, \quad (6)$$

since $\ddot{q}_k = F_k \cos \omega_k t$ and hence $\ddot{q}_k + \omega_k^2 q_k = F_k$, while q_k and $\dot{q}_k = F_k \sin(\omega_k t)/\omega_k$ both vanish at $t = 0$. Here $F_1 = F_0/\sqrt{3m}$ and $F_2 = -F_0/\sqrt{6m}$, so with $\omega_2 = 2\omega_1$ we find

$$\begin{aligned} u_1(t) &= \frac{F_0}{3k} (1 - \cos \omega_1 t) - \frac{F_0}{12k} (1 - \cos 2\omega_1 t), \\ u_2(t) &= \frac{F_0}{3k} (1 - \cos \omega_1 t) + \frac{F_0}{24k} (1 - \cos 2\omega_1 t). \end{aligned} \quad (7)$$

Averaging over the common period $2\pi/\omega_1$ replaces each factor $1 - \cos \omega_k t$ by unity, and gives $\bar{u}_1 = F_0/4k$ and $\bar{u}_2 = 3F_0/8k$; these are exactly the components of the static solution $\mathbf{K}^{-1}(0, F_0)^T$, as they should be. The response is the static deflection plus an undamped oscillation about it, and since $\omega_2 = 2\omega_1$ the motion here is periodic. As a consistency check on eq. (4), note that for $\text{Im } \omega < 0$ the transform of $H(t)$ is $1/(i\omega)$, while

$$\int_0^\infty \frac{1 - \cos \omega_k t}{\omega_k^2} e^{-i\omega t} dt = \frac{1}{\omega_k^2} \left(\frac{1}{i\omega} - \frac{i\omega}{\omega_k^2 - \omega^2} \right) = \frac{1}{i\omega(\omega_k^2 - \omega^2)}, \quad (8)$$

so the transform of eq. (6) is precisely $F_k \tilde{H}(\omega) / (\omega_k^2 - \omega^2)$, the k th coefficient in eq. (4). Everything in the lecture's derivation, including the $\frac{1 - \cos \omega_k t}{\omega_k^2}$ time dependence used in the second problem set, is already present in this two-dimensional model; the continuum case differs only in having infinitely many modes.

Example 2: A free-free rod. A uniform elastic rod of density ρ and Young's modulus E occupies $0 \leq x \leq L$ and has traction-free ends. Its longitudinal displacement $u(x, t)$ satisfies $\rho \ddot{u} - Eu'' = -\partial_x \bar{S}$ with $Eu' = \bar{S}$ at $x = 0, L$, where $\bar{S}(x, t)$ is a stress glut. (a) Write down the weak form and the normalised eigenfunctions $|k\rangle$ and eigenfrequencies ω_k . (b) Show that the rigid-translation mode $k = 0$ is not excited by any stress glut. (c) For the point source $\bar{S} = M_0 \delta(x - x_s) H(t)$ with $0 < x_s < L$, write down the response using the time-domain mode sum of the second problem set, check the initial conditions and, by direct substitution, the equation of motion, and interpret the long-time average of the solution.

Solution. (a) Multiplying the equation of motion by a test function w^* , integrating over the rod, and integrating by parts using the boundary condition $Eu' - \bar{S} = 0$ at the ends gives

$$\int_0^L \rho w^* \ddot{u} dx + \int_0^L E \frac{\partial w^*}{\partial x} \frac{\partial u}{\partial x} dx = \int_0^L \bar{S} \frac{\partial w^*}{\partial x} dx, \quad (9)$$

which is $\langle w|P|\ddot{u}\rangle + \langle w|H|u\rangle = \langle w|f\rangle$ in one dimension. Time-harmonic free solutions $s(x)e^{i\omega t}$ satisfy $Es'' = -\omega^2 \rho s$ with $s'(0) = s'(L) = 0$, so $s \propto \cos(k\pi x/L)$ with $k = 0, 1, 2, \dots$ and

$$\omega_k = \frac{k\pi c}{L}, \quad c = \sqrt{\frac{E}{\rho}}. \quad (10)$$

Since $\int_0^L \cos^2(k\pi x/L) dx = L/2$ for $k \geq 1$ and L for $k = 0$, the eigenfunctions normalised by $\langle k|P|k'\rangle = \int_0^L \rho |k\rangle |k'\rangle dx = \delta_{kk'}$ are

$$|0\rangle = \frac{1}{\sqrt{\rho L}}, \quad |k\rangle = \sqrt{\frac{2}{\rho L}} \cos \frac{k\pi x}{L} \quad (k \geq 1). \quad (11)$$

Each eigenspace is one-dimensional, so the second label m of the lecture is not needed.

(b) The force term for the k th mode is $\langle k|f\rangle = \int_0^L \bar{S} \partial_x |k\rangle dx$. For $k = 0$ the eigenfunction is constant, its derivative vanishes, and so $\langle 0|f\rangle = 0$ whatever $\bar{S}$ may be. This is the one-dimensional version of the lecture's statement that the trivial modes cannot be excited by a seismic source: an internal source exerts no net force, and the centre of mass, $\int_0^L \rho u dx \propto \langle 0|P|u\rangle$, remains at rest.

(c) For the point source, $\langle k|f\rangle(t) = M_0 H(t) \partial_x |k\rangle(x_s) = F_k H(t)$ with

$$F_k = -M_0 \sqrt{\frac{2}{\rho L}} \frac{k\pi}{L} \sin \frac{k\pi x_s}{L}. \quad (12)$$

The mode sum from the second problem set requires the time integrals $\int_0^t \sin[\omega_k(t -$

$t')]$ $H(t') dt' / \omega_k = (1 - \cos \omega_k t) / \omega_k^2$, and so

$$u(x, t) = \sum_{k=1}^{\infty} F_k \frac{1 - \cos \omega_k t}{\omega_k^2} |k\rangle(x) = -\frac{2M_0}{\pi E} \sum_{k=1}^{\infty} \frac{1}{k} \sin \frac{k\pi x_s}{L} \cos \frac{k\pi x}{L} \left(1 - \cos \frac{k\pi ct}{L}\right), \quad (13)$$

where we used $\omega_k^2 = Ek^2\pi^2/(\rho L^2)$ to simplify the coefficients. At $t = 0$ every term vanishes, and the time derivative brings down a factor $\omega_k \sin \omega_k t$ that also vanishes, so $u(x, 0) = \dot{u}(x, 0) = 0$ as required. The boundary conditions hold term by term, since $\partial_x \cos(k\pi x/L)$ vanishes at both ends, in agreement with $Eu' = \bar{S} = 0$ there.

For the direct substitution we apply $\rho \partial_t^2 - E \partial_x^2$ to each term of eq. (13). Since $E(k\pi/L)^2 = \rho \omega_k^2$, the k th term is multiplied by $\rho \omega_k^2 [\cos \omega_k t + (1 - \cos \omega_k t)] = \rho \omega_k^2$ for $t > 0$, so

$$\rho \ddot{u} - Eu'' = -\frac{2M_0}{\pi E} \sum_{k=1}^{\infty} \frac{\rho \omega_k^2}{k} \sin \frac{k\pi x_s}{L} \cos \frac{k\pi x}{L} = -\frac{2\pi M_0}{L^2} \sum_{k=1}^{\infty} k \sin \frac{k\pi x_s}{L} \cos \frac{k\pi x}{L}. \quad (14)$$

To identify the right-hand side we use the completeness relation, $u(x) = \sum_k \langle k|P|u\rangle |k\rangle$, which on writing out the inner product becomes $\rho \sum_k |k\rangle(x) |k\rangle(x') = \delta(x - x')$, and which, with eq. (11), is the cosine series

$$\delta(x - x_s) = \frac{1}{L} + \frac{2}{L} \sum_{k=1}^{\infty} \cos \frac{k\pi x_s}{L} \cos \frac{k\pi x}{L}. \quad (15)$$

Differentiating with respect to x_s , and using $\partial_{x_s} \delta(x - x_s) = -\delta'(x - x_s)$, gives

$$\delta'(x - x_s) = \frac{2\pi}{L^2} \sum_{k=1}^{\infty} k \sin \frac{k\pi x_s}{L} \cos \frac{k\pi x}{L}, \quad (16)$$

and comparison with eq. (14) shows that $\rho \ddot{u} - Eu'' = -M_0 \delta'(x - x_s) H(t) = -\partial_x \bar{S}$, as required. The series in eqs. (15) and (16) converge only in the sense of distributions, which is the sense in which a point source satisfies the equation of motion; the series in eq. (13) for the displacement itself has coefficients decaying like $1/k$ and converges, though slowly, to a function with a jump.

That jump is the physical content of the long-time average. Averaging over time replaces each $1 - \cos \omega_k t$ by 1, and the resulting series is the cosine series of the static solution. Indeed, the static problem $Eu' = \bar{S}$ integrates to $u = (M_0/E)H(x - x_s) + \text{const}$, and the constant is fixed by $\langle 0|P|u\rangle = 0$, i.e. zero mean displacement. The function $g(x) = (M_0/E)[H(x - x_s) - (L - x_s)/L]$ has $\int_0^L g dx = 0$ and cosine coefficients

$$\frac{2}{L} \int_0^L g(x) \cos \frac{k\pi x}{L} dx = \frac{2M_0}{EL} \int_{x_s}^L \cos \frac{k\pi x}{L} dx = -\frac{2M_0}{\pi Ek} \sin \frac{k\pi x_s}{L}, \quad (17)$$

which are exactly the time-averaged coefficients in eq. (13). The point stress glut is therefore a dislocation: the material on the two sides of x_s is permanently offset by M_0/E , the mean displacement stays zero, and the dynamic solution consists of the waves radiated by the sudden appearance of this offset, reflecting for ever between the free ends because we have neglected attenuation. Numerically, a partial sum with 20 000 terms reproduces the step to four decimal places away from x_s , and the mode sum for a Gaussian-smoothed glut agrees with a finite-difference solution of the rod to the discretisation error of the latter.

Example 3: Verifying the time-domain mode sum. The second problem set quotes the time-domain solution of the forced problem in a non-rotating earth model,

$$\mathbf{u}(\mathbf{x}, t) = \sum_{km} \left\{ \int_0^t \frac{\sin[\omega_k(t - t')]}{\omega_k} \int_M \bar{S}_{ij}(\mathbf{x}', t') \frac{\partial |km\rangle^*}{\partial x'_j}(\mathbf{x}') d^3\mathbf{x}' dt' \right\} |km\rangle(\mathbf{x}). \quad (18)$$

Show by direct substitution that this satisfies the weak form $\langle \mathbf{w} | P | \ddot{\mathbf{u}} \rangle + \langle \mathbf{w} | H | \mathbf{u} \rangle = \langle \mathbf{w} | \mathbf{f} \rangle$ for every test function $\mathbf{w}$, together with the initial conditions $\mathbf{u}(\mathbf{x}, 0) = \dot{\mathbf{u}}(\mathbf{x}, 0) = \mathbf{0}$. Then show that its Fourier transform is the frequency-domain eigenfunction expansion of the lecture, and specialise to a source that is a step in time, $\bar{S}_{ij}(\mathbf{x}, t) = \bar{S}_{ij}(\mathbf{x})H(t)$.

Solution. The inner spatial integral in eq. (18) is the force term of the lecture evaluated on the (k, m) th eigenfunction,

$$F_{km}(t) = \langle km | \mathbf{f} \rangle(t) = \int_M \bar{S}_{ij}(\mathbf{x}, t) \frac{\partial |km\rangle_i^*}{\partial x_j} d^3\mathbf{x}, \quad (19)$$

so that eq. (18) reads $\mathbf{u} = \sum_{km} q_{km}(t) |km\rangle$ with the modal coordinates

$$q_{km}(t) = \int_0^t \frac{\sin[\omega_k(t - t')]}{\omega_k} F_{km}(t') dt'. \quad (20)$$

A word about the trivial modes. If $\omega_k = 0$ then $\sin[\omega_k(t - t')]/\omega_k$ is to be read as its limit $t - t'$, but for these modes $|km\rangle_i = a_i + B_{ij}x_j$ with B_{ij} anti-symmetric, so $\partial_j |km\rangle_i = B_{ij}$, and its contraction with the symmetric $\bar{S}_{ij}$ in eq. (19) vanishes. The trivial modes therefore contribute nothing, and in what follows we may take $\omega_k > 0$.

Differentiating eq. (20) with respect to its upper limit and under the integral sign,

$$\dot{q}_{km}(t) = \frac{\sin 0}{\omega_k} F_{km}(t) + \int_0^t \cos[\omega_k(t - t')] F_{km}(t') dt' = \int_0^t \cos[\omega_k(t - t')] F_{km}(t') dt', \quad (21)$$

and differentiating once more,

$$\ddot{q}_{km}(t) = F_{km}(t) - \omega_k \int_0^t \sin[\omega_k(t - t')] F_{km}(t') dt' = F_{km}(t) - \omega_k^2 q_{km}(t). \quad (22)$$

Thus each modal coordinate satisfies the forced oscillator equation $\ddot{q}_{km} + \omega_k^2 q_{km} = F_{km}$, and from eqs. (20) and (21) both $q_{km}(0)$ and $\dot{q}_{km}(0)$ vanish, since the integrals are over an interval of zero length. Hence $\mathbf{u}(\mathbf{x}, 0) = \dot{\mathbf{u}}(\mathbf{x}, 0) = \mathbf{0}$. Eq. (20) is just Duhamel's formula for this oscillator.

Now substitute into the weak form, differentiating the sum term by term. Using the sesquilinearity of the forms and the eigenvalue equation $\langle \mathbf{w} | H | km \rangle = \omega_k^2 \langle \mathbf{w} | P | km \rangle$, which holds for every test function $\mathbf{w}$,

$$\begin{aligned} \langle \mathbf{w} | P | \ddot{\mathbf{u}} \rangle + \langle \mathbf{w} | H | \mathbf{u} \rangle &= \sum_{km} [\ddot{q}_{km} \langle \mathbf{w} | P | km \rangle + q_{km} \langle \mathbf{w} | H | km \rangle] \\ &= \sum_{km} (\ddot{q}_{km} + \omega_k^2 q_{km}) \langle \mathbf{w} | P | km \rangle = \sum_{km} F_{km} \langle \mathbf{w} | P | km \rangle. \end{aligned} \quad (23)$$

It remains to identify the last sum with $\langle \mathbf{w} | \mathbf{f} \rangle$. Expanding the test function using completeness, $\mathbf{w} = \sum_{km} \langle km | P | \mathbf{w} \rangle | km \rangle$, and recalling that $\langle \mathbf{w} | \mathbf{f} \rangle$ is conjugate-linear in $\mathbf{w}$, we have

$$\langle \mathbf{w} | \mathbf{f} \rangle = \sum_{km} \langle km | P | \mathbf{w} \rangle^* \langle km | \mathbf{f} \rangle = \sum_{km} \langle \mathbf{w} | P | km \rangle F_{km}, \quad (24)$$

where the self-adjointness of the kinetic energy form, $\langle km | P | \mathbf{w} \rangle^* = \langle \mathbf{w} | P | km \rangle$, was used in the last step. Eqs. (23) and (24) agree for every $\mathbf{w}$, so eq. (18) solves the weak form, and because the weak form incorporates the boundary conditions, the traction condition on ∂M is satisfied automatically by the eigenfunctions. Only three ingredients were needed: the eigenvalue equation, the orthonormality and completeness of the eigenfunctions, and the self-adjointness of P .

For the connection with the frequency domain, we take the transform of eq. (20), namely $\tilde{q}_{km}(\omega) = \int_0^\infty q_{km} e^{-i\omega t} dt$. Writing the exponential as $e^{-i\omega(t-t')} e^{-i\omega t'}$ and exchanging the order of integration over the region $0 \leq t' \leq t < \infty$ gives the causal convolution theorem $\tilde{q}_{km} = \tilde{g}_k \tilde{F}_{km}$ with

$$\tilde{g}_k(\omega) = \int_0^\infty \frac{\sin \omega_k t}{\omega_k} e^{-i\omega t} dt = \frac{1}{2i\omega_k} \left[\frac{1}{i(\omega - \omega_k)} - \frac{1}{i(\omega + \omega_k)} \right] = \frac{1}{\omega_k^2 - \omega^2}, \quad (25)$$

where the integral converges for $\text{Im } \omega < 0$, the standard device that corresponds physically to a slight attenuation moving the poles off the real axis. Hence

$$\tilde{\mathbf{u}} = \sum_{km} \frac{\langle km | \tilde{\mathbf{f}} \rangle}{\omega_k^2 - \omega^2} | km \rangle, \quad (26)$$

which is the eigenfunction expansion solution derived in the lecture; eq. (18) is its inverse transform, and the contour integration mentioned in the lecture simply evaluates the residues at $\omega = \pm \omega_k$ to produce $\sin(\omega_k t) / \omega_k$. Finally, for a step-in-time source $F_{km}(t) = F_{km} H(t)$ with $F_{km} = \int_M \bar{S}_{ij}(\mathbf{x}) \partial_j | km \rangle_i^* d^3 \mathbf{x}$ constant, the time integral in eq. (20) is elementary and

$$\mathbf{u}(\mathbf{x}, t) = \sum_{km} F_{km} \frac{1 - \cos \omega_k t}{\omega_k^2} | km \rangle(\mathbf{x}), \quad (27)$$

the form used in Examples 1 and 2. Each mode is then displaced to a new static equilibrium F_{km} / ω_k^2 about which it oscillates for ever with amplitude equal to that offset, which is why a step source excites the gravest modes, with the smallest ω_k^2 , so strongly.

Example 4: Spectral resolution and record length. A seismogram is recorded for a time T , which we model by the boxcar window $h(t) = 1$ for $0 \leq t \leq T$ and $h(t) = 0$ otherwise. (a) Compute $\tilde{h}(\omega)$, locate the first zero of $|\tilde{h}|$ and find the full width at half maximum (FWHM) of the peak, both for the power spectrum $|\tilde{h}|^2$ and the amplitude spectrum $|\tilde{h}|$. (b) Show that the observed spectrum of an undamped mode $u(t) = A \cos \omega_0 t$ is a copy of $\tilde{h}$ centred on ω_0 , and state the criterion for two modes with nearby frequencies to be resolved. (c) The five singlets of the multiplet ${}_0S_2$ are centred on $f_0 \approx 0.3094$ mHz and adjacent singlets are separated by about $4.6 \mu\text{Hz}$. Estimate the record length needed to resolve them, and compare it with the decay time of the mode, for which $Q \approx 510$.

Solution. (a) With the lecture's convention,

$$\tilde{h}(\omega) = \int_0^T e^{-i\omega t} dt = \frac{1 - e^{-i\omega T}}{i\omega} = T e^{-i\omega T/2} \frac{\sin(\omega T/2)}{\omega T/2}, \quad (28)$$

so $|\tilde{h}(\omega)| = T |\text{sinc}(\omega T/2)|$ with $\text{sinc } x = \sin x/x$. In terms of the frequency $f = \omega/2\pi$ the argument is $x = \pi f T$, and the zeros of $|\tilde{h}|$ lie at $f = n/T$ for $n = \pm 1, \pm 2, \dots$; the first zero is at $f = 1/T$. The power spectrum $|\tilde{h}|^2 = T^2 \text{sinc}^2 x$ falls to half its peak value where $\sin x/x = 1/\sqrt{2}$, which has the root $x_{1/2} = 1.3916$, and the full width of the peak is therefore $2x_{1/2}/\pi$ in units of $1/T$:

$$\Delta f_{\text{FWHM}}(|\tilde{h}|^2) = \frac{2x_{1/2}}{\pi T} = \frac{0.886}{T}. \quad (29)$$

For the amplitude spectrum the half-maximum condition is $\sin x/x = 1/2$, with root $x = 1.8955$, giving $\Delta f_{\text{FWHM}}(|\tilde{h}|) = 1.207/T$. Whichever definition is used, the spectral line has a width of order $1/T$, and the main lobe between the first zeros has total width $2/T$.

(b) A single undamped mode observed through the window has transform

$$\tilde{u}_{\text{obs}}(\omega) = \int_0^T A \cos(\omega_0 t) e^{-i\omega t} dt = \frac{A}{2} [\tilde{h}(\omega - \omega_0) + \tilde{h}(\omega + \omega_0)], \quad (30)$$

which is the convolution result of the lecture made explicit: the transform of the infinite record, with its poles at $\pm\omega_0$, is replaced by copies of $\tilde{h}$ centred on $\pm\omega_0$. Near $\omega \approx \omega_0 > 0$ the second term is smaller by a factor of order $1/(\omega_0 T)$ and can be neglected whenever the record contains many cycles (for $0S_2$ and $T = 24 \text{ h}$, $\omega_0 T \approx 170$). Two modes at f_1 and f_2 each contribute such a line, and by the usual Rayleigh-type criterion they are resolved when the peak of one lies no closer than the first zero of the other, i.e. when $|f_1 - f_2| \gtrsim 1/T$. Fig. 1 shows the amplitude spectrum of two equal-amplitude cosines separated by $4.6 \mu\text{Hz}$ for two record lengths.

(c) With adjacent singlets $4.6 \mu\text{Hz}$ apart the criterion of part (b) requires

$$T \gtrsim \frac{1}{4.6 \times 10^{-6} \text{ Hz}} = 2.2 \times 10^5 \text{ s} \approx 60 \text{ h}, \quad (31)$$

that is, two and a half days. A one-day record has $1/T = 11.6 \mu\text{Hz}$, larger than the width of the entire multiplet ($4 \times 4.6 \approx 18 \mu\text{Hz}$), and shows a single broad peak, while a five-day record has $1/T = 2.3 \mu\text{Hz}$ and separates the singlets clearly, as in Fig. 1.

Whether such records are available is set by attenuation. A mode with quality factor Q decays as $e^{-\omega_0 t/2Q} = e^{-\pi f_0 t/Q}$, so the amplitude e-folding time is

$$\tau = \frac{Q}{\pi f_0} = \frac{510}{\pi \times 3.094 \times 10^{-4} \text{ s}^{-1}} = 5.2 \times 10^5 \text{ s} \approx 6 \text{ days}. \quad (32)$$

This is comfortably longer than the 60 h required: after five days the amplitude has fallen only to $e^{-120/146} \approx 0.44$ of its initial value, so for a sufficiently large earthquake the signal remains above the noise for long enough. Attenuation also sets a floor on the achievable resolution, because the transform of $e^{-\pi f_0 t/Q} \cos \omega_0 t$ is a Lorentzian rather

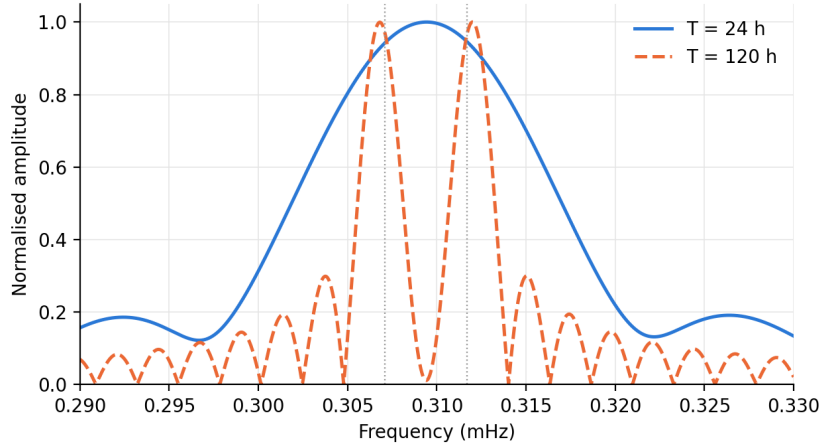


Fig. 1: Amplitude spectrum, normalised to its maximum, of the sum of two equal-amplitude cosines at the frequencies of two adjacent singlets of ${}_0S_2$ (dotted lines, separated by $4.6 \mu\text{Hz}$) observed through a boxcar window of length 24 h and 120 h. Attenuation is neglected.

than a delta function, with FWHM $f_0/Q \approx 0.6 \mu\text{Hz}$ in the power spectrum. Since this is well below the singlet spacing, the splitting of ${}_0S_2$ is resolvable, and it is indeed observed after the very largest earthquakes. Extending the record much beyond a few τ gains nothing, as the window then contains mostly noise. The situation deteriorates at higher frequencies, where f/Q grows to values comparable with the widths of the multiplets, and this is why individual peaks become hard to identify in the higher-frequency part of the spectrum shown in the lecture.

Example 5: Pointwise stability of an isotropic solid. In an isotropic solid without pre-stress the elastic tensor is $A_{ijkl} = \lambda \delta_{ij} \delta_{kl} + \mu (\delta_{ik} \delta_{jl} + \delta_{il} \delta_{jk})$. (a) For a complex symmetric tensor e_{ij} , decompose e_{ij} into isotropic and deviatoric parts and show that $A_{ijkl} e_{ij}^* e_{kl} = \kappa |e_{kk}|^2 + 2\mu d_{ij}^* d_{ij}$, where d_{ij} is the deviatoric part and $\kappa = \lambda + \frac{2}{3}\mu$. (b) Show that the pointwise stability condition $A_{ijkl} e_{ij}^* e_{kl} \geq c_0 e_{ij}^* e_{ij}$ holds for some $c_0 > 0$ if and only if $\kappa > 0$ and $\mu > 0$, and find the largest admissible c_0 . (c) Show that these conditions imply $\alpha > \beta > 0$ for the P- and S-wave speeds, but that the converse fails, using $\lambda = -0.9\mu$ as a counter-example. Explain physically why the two conditions differ.

Solution. (a) Write e_{ij} as the sum of an isotropic part proportional to the identity and a traceless remainder,

$$e_{ij} = \frac{1}{3} \theta \delta_{ij} + d_{ij}, \quad \theta = e_{kk}, \quad d_{kk} = 0. \quad (33)$$

The two parts are orthogonal: $\delta_{ij} d_{ij} = d_{kk} = 0$, so that

$$e_{ij}^* e_{ij} = \frac{1}{9} |\theta|^2 \delta_{ij} \delta_{ij} + d_{ij}^* d_{ij} + \frac{1}{3} (\theta^* d_{kk} + \theta d_{kk}^*) = \frac{1}{3} |\theta|^2 + d_{ij}^* d_{ij}. \quad (34)$$

Contracting the elastic tensor with $e_{ij}^* e_{kl}$, the λ term gives $\lambda e_{ii}^* e_{kk} = \lambda |\theta|^2$, and the two μ terms give $\mu (e_{ij}^* e_{ij} + e_{ji}^* e_{ji}) = 2\mu e_{ij}^* e_{ij}$ by the symmetry of e_{ij} . Using eq. (34),

$$A_{ijkl} e_{ij}^* e_{kl} = \lambda |\theta|^2 + 2\mu \left(\frac{1}{3} |\theta|^2 + d_{ij}^* d_{ij} \right) = \kappa |\theta|^2 + 2\mu d_{ij}^* d_{ij}, \quad (35)$$

with $\kappa = \lambda + \frac{2}{3}\mu$ the bulk modulus. The strain energy density $\frac{1}{2}A_{ijkl}e_{ij}e_{kl} = \frac{1}{2}\kappa\theta^2 + \mu d_{ij}d_{ij}$ thus splits into a part that resists change of volume, governed by κ , and a part that resists change of shape, governed by μ .

(b) Combining eqs. (34) and (35),

$$A_{ijkl}e_{ij}^*e_{kl} - c_0e_{ij}^*e_{ij} = (3\kappa - c_0)\frac{1}{3}|\theta|^2 + (2\mu - c_0)d_{ij}^*d_{ij}. \quad (36)$$

If $c_0 \leq 3\kappa$ and $c_0 \leq 2\mu$ the right-hand side is non-negative for every symmetric e_{ij} . Conversely, the choice $e_{ij} = \delta_{ij}$ (a pure dilatation, $\theta = 3$, $d_{ij} = 0$) makes the right-hand side $3(3\kappa - c_0)$, and a traceless choice such as $e_{12} = e_{21} = 1$ with all other components zero makes it $2(2\mu - c_0)$, so both inequalities are also necessary. Pointwise stability with some $c_0 > 0$ therefore holds if and only if $\kappa > 0$ and $\mu > 0$, and the largest admissible constant is

$$c_0 = \min(3\kappa, 2\mu). \quad (37)$$

In other words, regarded as a symmetric linear map on the six-dimensional space of symmetric tensors, A_{ijkl} has the eigenvalue 3κ on the one-dimensional space of isotropic tensors and the eigenvalue 2μ on the five-dimensional space of deviatoric tensors, and c_0 is its smallest eigenvalue.

(c) From the Christoffel equation in an isotropic material, $\rho\alpha^2 = \lambda + 2\mu = \kappa + \frac{4}{3}\mu$ and $\rho\beta^2 = \mu$. If $\kappa > 0$ and $\mu > 0$ then $\beta^2 > 0$ and $\rho(\alpha^2 - \beta^2) = \kappa + \frac{1}{3}\mu > 0$, so $\alpha > \beta > 0$. The converse fails: real wave speeds with $\alpha > \beta$ require only $\mu > 0$ and $\lambda > -2\mu$, whereas $\kappa > 0$ requires the stronger $\lambda > -\frac{2}{3}\mu$. For $\lambda = -0.9\mu$ with $\mu > 0$ we have

$$\kappa = -0.9\mu + \frac{2}{3}\mu = -\frac{7}{30}\mu < 0, \quad \rho\alpha^2 = 1.1\mu > 0, \quad \alpha = \sqrt{1.1}\beta \approx 1.05\beta, \quad (38)$$

so P- and S-waves propagate with real speeds and $\alpha > \beta > 0$, yet the uniform dilatation $e_{ij} = \delta_{ij}$ has $A_{ijkl}e_{ij}e_{kl} = 9\kappa = -2.1\mu < 0$ and the material is not pointwise stable.

The reason is that plane waves test the elastic tensor only on strains of a very special kind. A plane wave with polarisation $\mathbf{a}$ and unit propagation direction $\hat{\mathbf{p}}$ produces the rank-one strain $e_{ij} = \frac{1}{2}(a_i\hat{p}_j + a_j\hat{p}_i)$, for which, using the symmetries of A_{ijkl} , $A_{ijkl}e_{ij}e_{kl} = A_{ijkl}a_i\hat{p}_ja_k\hat{p}_l = \rho a_i\Gamma_{ik}(\hat{\mathbf{p}})a_k$ with Γ_{ik} the Christoffel matrix. Such a strain has $\theta = \mathbf{a} \cdot \hat{\mathbf{p}}$ and $e_{ij}e_{ij} = \frac{1}{2}[\|\mathbf{a}\|^2 + (\mathbf{a} \cdot \hat{\mathbf{p}})^2]$, so by eq. (34) its deviatoric part satisfies $d_{ij}d_{ij} = \frac{1}{2}\|\mathbf{a}\|^2 + \frac{1}{6}(\mathbf{a} \cdot \hat{\mathbf{p}})^2$, and eq. (35) gives

$$A_{ijkl}e_{ij}e_{kl} = \mu\|\mathbf{a}\|^2 + \left(\kappa + \frac{1}{3}\mu\right)(\mathbf{a} \cdot \hat{\mathbf{p}})^2 = \rho \left[\beta^2\|\mathbf{a}\|^2 + (\alpha^2 - \beta^2)(\mathbf{a} \cdot \hat{\mathbf{p}})^2\right], \quad (39)$$

in agreement with the Christoffel matrix $\Gamma_{ik} = \alpha^2\hat{p}_i\hat{p}_k + \beta^2(\delta_{ik} - \hat{p}_i\hat{p}_k)$ of Lecture 15. Even the longitudinal case $\mathbf{a} = \hat{\mathbf{p}}$ has $d_{ij}d_{ij} = \frac{2}{3}$ against $\frac{1}{3}\theta^2 = \frac{1}{3}$: a rank-one strain is never a pure dilatation, and its deviatoric part always outweighs its isotropic part. A sufficiently positive shear modulus can therefore mask a negative bulk modulus in eq. (39), which stays positive so long as $\kappa > -\frac{4}{3}\mu$. A homogeneous compression, by contrast, has no deviatoric part at all; it is not a plane wave of any finite wavelength, and it is precisely the strain that a negative κ makes energetically favourable.

The consequence for a finite body is decisive. Take $u_i = x_i$ on any body M made of the material in eq. (38), so that $e_{ij} = \delta_{ij}$ everywhere and, in the non-gravitating case,

$\langle \mathbf{u} | H | \mathbf{u} \rangle = \int_M 9\kappa d^3\mathbf{x} < 0$. Expanding $\mathbf{u} = \sum_{km} c_{km} |km\rangle$ with $c_{km} = \langle km | P | \mathbf{u} \rangle$, the eigenvalue equation with test function $\mathbf{u}$ and the self-adjointness of P give

$$\langle \mathbf{u} | H | \mathbf{u} \rangle = \sum_{km} c_{km} \langle \mathbf{u} | H | km \rangle = \sum_{km} \omega_k^2 c_{km} \langle \mathbf{u} | P | km \rangle = \sum_{km} \omega_k^2 |c_{km}|^2, \quad (40)$$

which can only be negative if at least one squared eigenfrequency is negative. The body therefore has a mode that grows exponentially in time and is unstable, even though plane waves travel through the same material perfectly well. Positive wave speeds are a condition on the propagation of disturbances of finite wavelength; point-wise stability is the stronger condition that every strain, including a uniform change of volume, costs energy, and it is the latter that the lecture needs to conclude that all the squared eigenfrequencies of a non-gravitating earth model are non-negative.